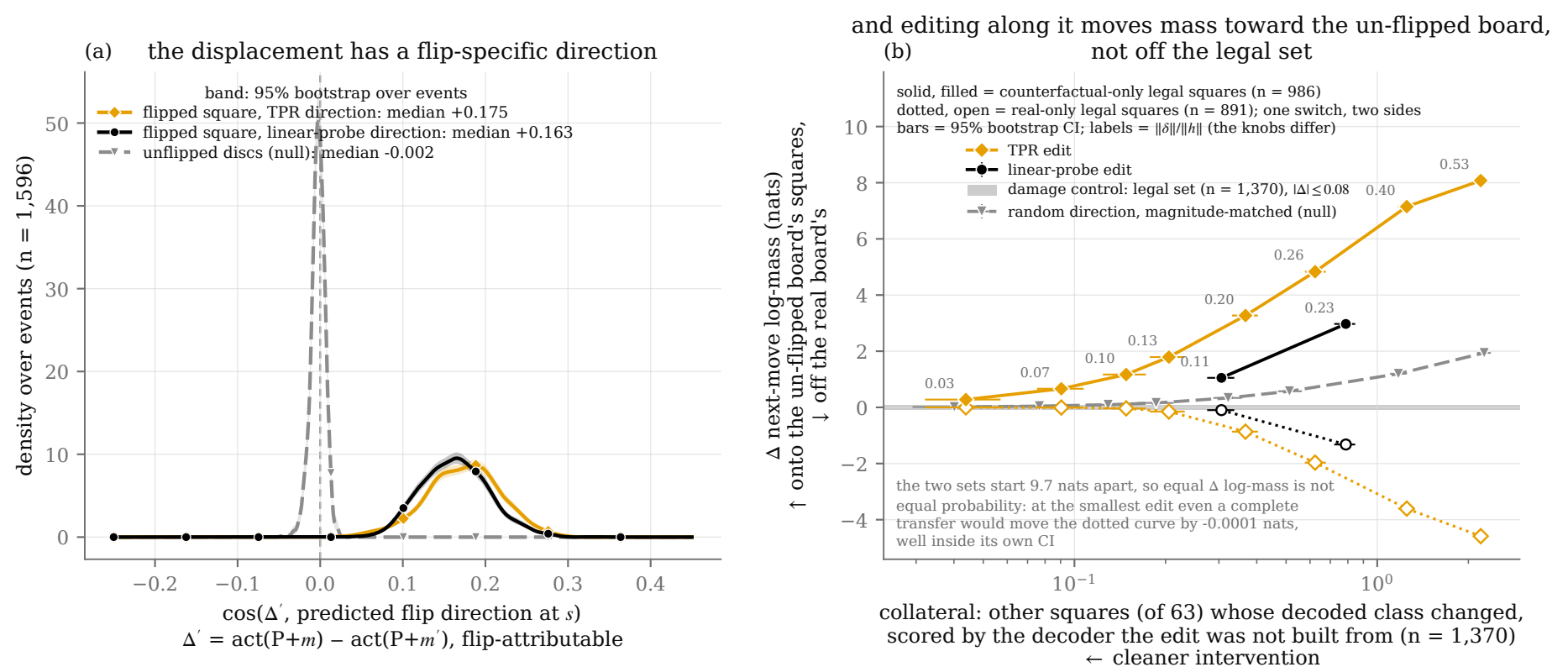
(a) the displacement has a flip-specific direction
band: 95% bootstrap over events
flipped square, TPR direction: median +0.175
flipped square, linear-probe direction: median +0.163
unflipped discs (null): median -0.002
density over events (n = 1,596)
cos(Δ′, predicted flip direction at s)
Δ′ = act(P+m) − act(P+m′), flip-attributable
and editing along it moves mass toward the un-flipped board,
(b) not off the legal set
solid, filled = counterfactual-only legal squares (n = 986)
dotted, open = real-only legal squares (n = 891); one switch, two sides
bars = 95% bootstrap CI; labels = ‖δ‖/‖h‖ (the knobs differ)
TPR edit
linear-probe edit
damage control: legal set (n = 1,370), |Δ| ≤ 0.08
random direction, magnitude-matched (null)
0.03
0.07
0.10
0.13
0.11
0.20
0.23
0.26
0.40
0.53
Δ next-move log-mass (nats)
↑ onto the un-flipped board's squares,
↓ off the real board's
the two sets start 9.7 nats apart, so equal Δ log-mass is not
equal probability: at the smallest edit even a complete
transfer would move the dotted curve by -0.0001 nats,
well inside its own CI
collateral: other squares (of 63) whose decoded class changed,
scored by the decoder the edit was not built from (n = 1,370)
← cleaner intervention